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from flask import Flask, render_template, request, jsonify
import pickle
import numpy as np

app = Flask(__name__)

# Load trained model
model = pickle.load(open('model/rainfall_model.pkl', 'rb'))

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/predict', methods=['POST'])
def predict():
    data = request.get_json()

    temperature = float(data['temperature'])
    humidity = float(data['humidity'])
    wind_speed = float(data['wind_speed'])
    pressure = float(data['pressure'])

    features = np.array([[temperature, humidity, wind_speed, pressure]])
    prediction = model.predict(features)

    return jsonify({
        'prediction': round(float(prediction[0]), 2)
    })

if __name__ == "__main__":
    app.run(debug=True)
```